

## OUTPUTS

1

```
Action: Attack
Next State: Middle
Reward: 10

Current State: Middle
Action: Attack
Next State: Winning
Reward: 50

Current State: Winning
Action: Finish
Next State: Goal
Reward: 100

Goal Reached!
Total Reward = 160
```

2

```
Q Table
Kitchen {'Move': 6.198389053344727}
Hall {'Move': 7.999820709228516}
Bedroom {'Move': 9.999990463256836}
Charging {'Move': 0}
```

3

```
Warehouse Robot

Current State: Storage
Next State: Pickup

Current State: Pickup
Next State: Delivery

Current State: Delivery
Next State: Goal

Goal Reached
Total Reward = 170
```

4

```
State Values
Goal = 100
C = 89.0
B = 79.1
A = 70.19
```

5

```
Advertisement Results

Ad1 Average Reward = 4.92 Selected 50 times
Ad2 Average Reward = 10.02 Selected 438 times
Ad3 Average Reward = 7.67 Selected 12 times

Best Advertisement: Ad2
```

6

```
Next State: 0
Reward: 0
-----
Step: 3
State: 0
Action: 2
Next State: 1
Reward: 0
-----
Step: 4
State: 1
Action: 1
Next State: 5
Reward: 0
-----
Episode Finished
```

7

```
Autonomous Robot Navigation
Step: 1
State: 0
Action: 1
Next State: 4
Reward: 0
-----
Step: 2
State: 4
Action: 2
Next State: 5
Reward: 0
-----
Episode Finished
```

8

```
Minimum Cost from Each Node
A : 11
B : 7
C : 10
D : 2
E : 3
F : 0

Optimal Cost from A to F = 11
```

9

```
Living Room Reward = 9
Kitchen Reward = 5
Bedroom Reward = 7
Bathroom Reward = 4

Episode 10
Living Room Reward = 7
Kitchen Reward = 10
Bedroom Reward = 5
Bathroom Reward = 7

Estimated State Values
Living Room : 5.6
Kitchen : 5.8
Bedroom : 6.0
Bathroom : 6.2
```

10

```
Episode 1
Low Load -> Medium Load Value = 2.5
Medium Load -> High Load Value = 5.0
Episode 2
Low Load -> Medium Load Value = 6.0
Medium Load -> High Load Value = 7.5
Episode 3
Low Load -> Medium Load Value = 8.88
Medium Load -> High Load Value = 8.75
Episode 4
Low Load -> Medium Load Value = 10.88
Medium Load -> High Load Value = 9.38
Episode 5
Low Load -> Medium Load Value = 12.16
Medium Load -> High Load Value = 9.69
Episode 6
```

11

```
Final Q Values
Warehouse = 22.8
Road = 19.98
Customer = 0
```

12

```
Epoch 7
Weight 1 = 0.85
Weight 2 = 1.08

Epoch 8
Weight 1 = 0.9
Weight 2 = 1.12

Epoch 9
Weight 1 = 0.95
Weight 2 = 1.16

Epoch 10
Weight 1 = 1.0
Weight 2 = 1.2
```

13

```
Episode 6
Policy Probability = 0.61

Episode 7
Policy Probability = 0.64

Episode 8
Policy Probability = 0.68

Episode 9
Policy Probability = 0.73

Episode 10
Policy Probability = 0.78
```

14

```
Second Floor -> Top Floor Value = 19.84
Ground Floor -> First Floor Value = 27.52
First Floor -> Second Floor Value = 27.33
Second Floor -> Top Floor Value = 19.92
Ground Floor -> First Floor Value = 28.56
First Floor -> Second Floor Value = 27.63
Second Floor -> Top Floor Value = 19.96
Ground Floor -> First Floor Value = 29.21
First Floor -> Second Floor Value = 27.8
Second Floor -> Top Floor Value = 19.98

Final State Values
Ground Floor = 29.21
First Floor = 27.8
Second Floor = 19.98
Top Floor = 0
```

15

```
Episode 6
Policy = 0.71

Episode 7
Policy = 0.78

Episode 8
Policy = 0.86

Episode 9
Policy = 0.95

Episode 10
Policy = 1.05

Training Completed
```

16

```
Episode 6
Lane Keeping Probability = 0.57

Episode 7
Lane Keeping Probability = 0.62

Episode 8
Lane Keeping Probability = 0.69

Episode 9
Lane Keeping Probability = 0.76

Episode 10
Lane Keeping Probability = 0.84

Training Finished
```

17

```
Hierarchical Reinforcement Learning

Performing Task: Clean Room
Reward: 20

Performing Task: Wash Dishes
Reward: 15

Performing Task: Arrange Books
Reward: 10

All Tasks Completed
Total Reward = 45
```

18

```
Meta Reinforcement Learning

Learning Task: Assembly
Knowledge Level = 5.0

Learning Task: Painting
Knowledge Level = 10.0

Learning Task: Packing
Knowledge Level = 15.0

Final Knowledge = 15.0
```

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### Multi-Agent Reinforcement Learning

Robot A -> Pick Item  
Reward = 10

Robot B -> Move Item  
Reward = 10

Robot C -> Store Item  
Reward = 10

All Robots Completed Their Tasks  
Total Reward = 30

20

### Inverse Reinforcement Learning

Walking Observed Reward = 5  
Running Observed Reward = 10  
Sitting Observed Reward = 2  
Standing Observed Reward = 3

Learned Optimal Activity: Running  
Maximum Reward = 10

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### Model-Based Reinforcement Learning

Current State: Start  
Next State: Room A  
Reward: 5

Current State: Room A  
Next State: Room B  
Reward: 10

Current State: Room B  
Next State: Goal  
Reward: 20

Goal Reached  
Total Reward = 35

22

```
Treatment : Medicine B
Reward : 90

Patient : Patient3
Treatment : Medicine C
Reward : 75

Patient : Patient4
Treatment : Medicine B
Reward : 95

Best Treatment Recommendation
Patient : Patient4
Treatment : Medicine B
Maximum Reward : 95
```

23

```
Traffic Signal Status

North Road Vehicles = 25
South Road Vehicles = 18
East Road Vehicles = 30
West Road Vehicles = 12

Green Signal Given To: East
Vehicles Cleared = 30
```

24

```
REINFORCE Trading System

Day1
Profit = 100
Policy = 0.55

Day2
Profit = 150
Policy = 0.62

Day3
Profit = 120
Policy = 0.69

Day4
Profit = 180
```



25

```
Advertisement Recommendation Results

Epsilon-Greedy CTR = 78 %
UCB CTR = 85 %
Thompson Sampling CTR = 91 %

Best Algorithm = Thompson Sampling
Highest CTR = 91 %
```

26

```
Dynamic Pricing Results

Epsilon-Greedy Revenue = Rs. 12000
UCB Revenue = Rs. 13500
Thompson Sampling Revenue = Rs. 14250

Best Pricing Strategy = Thompson Sampling
Maximum Revenue = Rs. 14250
```

27

```
Location : Junction
Action   : Turn Left
Reward  : 10

Location : Signal
Action   : Stop and Go
Reward  : 15

Location : Destination
Action   : Reached
Reward  : 20

Destination Reached Successfully
Total Reward = 50
```

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```
Optimal State Values

A = 70.19
B = 79.1
C = 89.0
Goal = 100

Optimal Path
A -> B -> C -> Goal
```

29

```
Traffic Signal Optimization

North Vehicles = 30
South Vehicles = 18
East Vehicles = 25
West Vehicles = 12

Optimal Green Signal : North
Maximum Vehicles Cleared : 30
```

30

```
Highway Weight = 0.85
Traffic Weight = 1.08

Epoch 8
Highway Weight = 0.9
Traffic Weight = 1.12

Epoch 9
Highway Weight = 0.95
Traffic Weight = 1.16

Epoch 10
Highway Weight = 1.0
Traffic Weight = 1.2

Training Completed
```

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```
Computer Move -> Win Q = 14.88
Start -> Player Move Q = 22.32
Player Move -> Computer Move Q = 20.99
Computer Move -> Win Q = 14.94
Start -> Player Move Q = 23.11
Player Move -> Computer Move Q = 21.22
Computer Move -> Win Q = 14.97
Start -> Player Move Q = 23.6
Player Move -> Computer Move Q = 21.35
Computer Move -> Win Q = 14.99

Final Q Values
Start = 23.6
Player Move = 21.35
Computer Move = 14.99
Win = 0
```

32

```
Episode 8  
Value = 0.9  
Advantage = 0.62  
Q Value = 1.52  
  
Episode 9  
Value = 0.95  
Advantage = 0.66  
Q Value = 1.61  
  
Episode 10  
Value = 1.0  
Advantage = 0.7  
Q Value = 1.7  
  
Training Completed
```

33

```
Episode 8  
Resources = 80  
Units Built = 16  
Reward = 432  
  
Episode 9  
Resources = 90  
Units Built = 18  
Reward = 540  
  
Episode 10  
Resources = 100  
Units Built = 20  
Reward = 660  
  
Training Completed
```

34

```
Temperature = 23  
Policy = 0.85  
  
Day 8  
Temperature = 22  
Policy = 0.9  
  
Day 9  
Temperature = 23  
Policy = 0.95  
  
Day 10  
Temperature = 22  
Policy = 1.0  
  
Optimization Completed
```

35

```
Investment Portfolio Analysis  
  
Portfolio A Predicted Value = Rs. 120000  
Portfolio B Predicted Value = Rs. 145000  
Portfolio C Predicted Value = Rs. 138000  
  
Best Portfolio = Portfolio B  
Highest Predicted Value = Rs. 145000
```

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```
MAXQ Hierarchical Reinforcement Learning  
  
Task : Navigate  
Reward : 10  
  
Task : Pick Object  
Reward : 15  
  
Task : Deliver Object  
Reward : 20  
  
All Tasks Completed  
Total Reward = 45
```

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```
Agent : Agent A
Task : Search Area
Reward: 10

Agent : Agent B
Task : Collect Object
Reward: 15

Agent : Agent C
Task : Deliver Object
Reward: 20

All Agents Completed Their Tasks
Total Reward = 45
Best Performing Agent = Agent C
```

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```
POMDP Navigation

State : Start
Belief = 0.65

State : Unknown Area
Belief = 0.8

State : Target
Belief = 0.95

Target Reached
```

39

```
Healthcare Management

Patient1
Resources Allocated = 5

Patient2
Resources Allocated = 3

Patient3
Resources Allocated = 4

Total Reward = 120
```

```
Personalized Learning

Student1
Learning Score = 70

Student2
Learning Score = 82

Student3
Learning Score = 91

Best Learning Performance
Student = Student3
Score = 91
```